

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2023-24

Zoology

Paper: ZOB-401 — Fundamental Endocrinology and Economic Zoology

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Fundamental Endocrinology) (Marks: 35)

Question 1. Answer the following parts:

(a) Justify whether the following statements are True or False: [4 × 2 = 8]

- (i) Insulin promotes synthesis of fatty acids in the liver and inhibits lipolysis.
- (ii) Chief cells of the parathyroid gland are only sensitive to calcium conjugated with anions like sulphates and citrates.
- (iii) Thyroid hormone increases BMR of all the tissues present in the body.
- (iv) Beer consumption causes frequent urination by suppressing oxytocin.

(b) Define the following: [3 × 1 = 3]

- (i) Paracrine mode of delivery
- (ii) Islets of Langerhans
- (iii) LH-surge.

Question 2. Discuss the role of pituitary hormones. [12]

Question 3. Describe the differential role of parathormone in calcium and phosphate homeostasis. [12]

Question 4. Write short notes on the following: [3 × 4 = 12]

- (a) Physiological functions of testosterone
- (b) Renin-angiotensin-aldosterone system
- (c) Positive feedback mechanism.

SECTION – B
(Economic Zoology) (Marks: 35)

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answer in 2–3 sentences each: [3 × 2 = 6]

- (i) Chalkbrood disease in honeybee is caused by protozoan.
- (ii) Fibroin is a major glycoprotein of silk.
- (iii) Hackle and cape feathers of hen usually have pointed ends.

- (b) Define the following terms: [5 × 1 = 5]

- (i) Fish seed
- (ii) Culling
- (iii) Stick lac
- (iv) Bee propolis
- (v) Raceway culture.

Question 2. Answer the following parts:

- (a) Briefly discuss the advantages and disadvantages of cage poultry rearing method. [6]
- (b) Describe the process of graft tissue preparation and nucleus implantation in oysters for pearl formation. [6]

Question 3. Answer the following parts:

- (a) Discuss about different diseases of a cattle caused by bacteria with emphasis on symptoms and treatments. [6]
- (b) What are different types of silk? Briefly elaborate the different stages in the mulberry silk production. [6]

Question 4. Write short notes on the following: [3 × 4 = 12]

- (a) Nuptial flight
- (b) Fish by-products
- (c) Preservation methods of prawn.

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